

Backend & Supabase Setup Guide

Project Role

This backend module handles:

- Supabase database connection
 - Face data storage
 - Face matching
 - Duplicate vote prevention
 - Vote status update
-

Required Software

1. Python Version

Install:

Python 3.10

IMPORTANT: Do NOT use Python 3.13 because face_recognition and dlib may cause errors.

2. Install Required Libraries

Open terminal inside project folder and run:

```
pip install opencv-python face-recognition numpy supabase
```

Required Files

Keep all these files in the SAME folder:

```
face_utils.py  
store_face.py
```

```
match_face.py
db.py
```

Supabase Setup

Required Tables

Table 1: user

Column Name	Data Type
id	int8
name	text
face_encoding	text
has_voted	bool

Table 2: votes

Column Name	Data Type
id	int8
user_id	int8
candidate	text
created_at	timestamp

Add Supabase Credentials

In all backend files, replace:

```
url = "YOUR_SUPABASE_URL"
key = "YOUR_SUPABASE_KEY"
```

with actual Supabase project credentials.

Camera Notes

If laptop camera does not work:

Change:

```
cv2.VideoCapture(0)
```

To:

```
cv2.VideoCapture(1)
```

for external USB camera.

How To Register New User

Run:

```
python store_face.py
```

Steps: 1. Enter user name 2. Camera opens 3. Press ENTER 4. Face gets stored in Supabase

Expected Output:

```
Supabase connected ✓  
Enter user name:  
Press ENTER to capture face  
Face detected: 1  
Encoding generated ✓  
Face stored in database ✓
```

How To Match Face

Run:

```
python match_face.py
```

Expected Outputs:

If user has not voted:

Match found ✓ : Rohit
Allowed to vote ✓
Vote status updated ✓

If user already voted:

Match found ✓ : Sri Sai
Already voted ✗

Project Flow

```
Camera
  ↓
Face Encoding
  ↓
Fetch Users From Supabase
  ↓
Face Matching
  ↓
Check Voting Status
  ↓
Update Vote Status
```

Important Notes

- Use good lighting during face capture
 - Keep face clearly visible
 - Do not move while capturing
 - All users should register their own faces
 - Do not manually change names for another person's face
-

Common Errors & Fixes

Error: Camera not accessible

Fix:

Change camera index:

```
cv2.VideoCapture(1)
```

Error: No module named 'supabase'

Fix:

```
pip install supabase
```

Error: Face not detected

Fix:

- Improve lighting
 - Sit properly in front of camera
 - Ensure webcam is working
-

Error: Connection issue

Check:

- Internet connection
 - Supabase URL
 - Supabase key
-

Backend Features Completed

✅ Face Registration ✅ Face Matching ✅ Duplicate Vote Prevention ✅ Supabase Integration ✅ User Data Storage ✅ Vote Status Update ✅ Database Fetch Operations